A Novel Feature Construction Method for Tobacco Chlorophyll Estimation Based on Integral of UAV-Borne Hyperspectral Reflectance Curve

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***Abstract*— Chlorophyll, a key pigment in leaf photosynthesis, is crucial for monitoring tobacco growth, evaluating quality, and determining optimal harvest timing. Hyperspectral remote sensing (HRS) via unmanned aerial vehicles (UAVs) provides a viable method to assess tobacco leaf chlorophyll content (LCC) due to its real-time and high-throughput capabilities. However, the existing spectral feature extraction methods often suffer from variability due to external factors or internal parameters, resulting in unstable results. To address this, we proposed a novel feature construction method, termed segmented fitting for integral (SFI). This method divides the entire spectral curve into five areas based on the spectral response properties of chlorophyll and nitrogen and then selects a suitable fitting function for each area to calculate the integrals. This idea not only fully utilizes the spectral reflectance and curve shape characteristics but also remains largely unaffected by external factors and internal parameters. To further verify the stability and predictive capability of the SFI method, we compared it against three different feature extraction methods, including the successful projections algorithm (SPA), the recursive feature elimination (RFE), and the principal component analysis (PCA), and also with two ensemble learning-based modeling approaches, namely, random forest (RF) and adaptive boosting (AdaBoost). The results demonstrated that the SFI method can effectively reduce data dimensionality and enhance model performance. Finally, we applied the best-performing SFI-AdaBoost model in field prediction to generate chlorophyll distribution and error maps, which closely aligned with actual chlorophyll measurements and demonstrated its potential for practical evaluation of tobacco LCC.**

***Index Terms*— Chlorophyll, feature construction, hyperspectral, integral, segmented fitting, tobacco.**

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I. Introduction

TOBACCO is a pivotal economic crop in Southwest China, underpinning the local agricultural sector and significantly contributing to rural economic development. This crop not only forms a vital pillar of the regional economy but also provides a stable income for local farmers [1], [2]. In the marketplace, the value of tobacco leaves is directly linked to their quality grade after curing [3], [4]. Therefore, stabilizing or enhancing the quality grade of cured tobacco leaves is of significant importance to ensure the livelihoods of local farmers [5]. To achieve this, it is particularly important to select the appropriate fresh tobacco leaves for curing. Consequently, a crucial question arises: how can we determine whether these tobacco leaves are suitable for harvesting and curing?

Chlorophyll, the key pigment in leaf photosynthesis, is essential for harvesting solar radiation and converting it into chemical energy [6], [7]. In agriculture, the leaf chlorophyll content (LCC) of crops directly reflects their health, significantly influencing yield and quality [8], [9]. This is also true for tobacco, a typical broad-leaf crop [10]. On the one hand, LCC largely determines the coloration of tobacco leaves, which farmers use to assess leaf maturity and make harvesting decisions. On the other hand, the aromatic hydrocarbon precursors responsible for the characteristic aroma of cured tobacco are derived from chlorophyll degradation after curing [11], [12], [13]. However, excessive LCC can lead to incomplete chlorophyll degradation, thus diminishing the quality of tobacco leaves and adversely impacting the interests of farmers. Therefore, it is necessary to control the LCC of tobacco leaves used for curing within a reasonable range. In summary, LCC is an excellent indicator for assessing the growth status of tobacco leaves, determining the optimal harvesting time, and assessing the final quality grade. Accurate and real-time acquisition of field chlorophyll information is of great significance for modern tobacco production [14].

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Hyperspectral remote sensing (HRS) is an advanced technology for acquiring surface information based on the principles of electromagnetic radiation [15]. With its capability to obtain crop spectral information nondestructively, HRS techniques are widely applied in various agricultural activities, such as land use monitoring [16], [17], crop phenotyping [18], [19], and yield estimation [20], [21]. Moreover, to accommodate the needs of large-scale observation, unmanned aerial vehicles (UAVs) have been integrated into near-ground remote sensing and are widely used for high-throughput phenotyping [22], [23]. Therefore, this article aims to explore a feasible and efficient method for acquiring field-scale tobacco LCC information using UAV-borne HRS.

According to the existing literature, the most commonly used HRS-based approaches for LCC estimation are mainly physical models (PMs) [24], [25] and empirical models (EMs) [26], [27]. PMs seek to simulate the spectral reflection and absorption characteristics of chlorophyll across different wavelengths. These models take into account multiple factors, such as leaf surface structure [28], [29], soil background effects [30], [31], and atmospheric and lighting conditions [32]. Each factor is potentially described by one or multiple parameters, adding significant complexity to the modeling process. In contrast, EMs establish relationships by statistically analyzing the correlation between hyperspectral data and LCC. These models often utilize the reflectance data of specific bands or certain vegetation indices (VIs) as inputs, aiming to find empirical relationships between the data [33], [34], which is simple, efficient, and more suitable for practical applications. Nonetheless, EMs still face the problem of spectral feature construction. Most conventional VI-based EMs are sensitive to soil background or canopy structure [35], [36]. The related models also suffer from low robustness and are prone to saturation [15], [37]. In our view, VI-based models are relatively simple and easy to calculate, because they are usually constructed by combining two or three different spectral bands [34], [38]. However, considering the utilization of the entire spectrum, VI-based models are more suitable for tasks based on multispectral data rather than hyperspectral data [39], [40].

To improve spectral utilization while reducing hyperspectral data redundancy, various machine learning-based feature construction methods have been introduced, such as successful projections algorithm (SPA) [41], [42], genetic algorithm (GA) [43], and competitive adaptive reweighted sampling (CARS) [44]. These methods can screen out wavelengths that are sensitive to changes in LCC. By leveraging machine learning algorithms, quantitative inversion models can be established between these sensitive bands and LCC data, and often with excellent modeling performance. However, the selection of sensitive bands heavily depends on specific parameters and the choice of internal evaluation model. For example, within SPA, multiple linear regression is used to evaluate the sensitivity of each band. As a result, the sensitive bands identified may vary with different parameter sets, leading to variable modeling performance.

To accurately capture chlorophyll-sensitive bands, some studies have utilized the spectral response of chlorophyll to construct relevant features. Due to the electronic transitions, chlorophyll primarily absorbs blue (about 400–500 nm) light and red (about 600–700 nm) light in the spectrum, while primarily reflecting green (about 500–600 nm) light [45]. Furthermore, as an important component of chlorophyll, nitrogen can also be utilized to characterize the LCC, and the 750–920 nm shows a high correlation with leaf nitrogen content [46]. Based on these properties, Song et al. [47] introduced a secondary optimization feature extraction method. The backward interval partial least-squares method was first used to screen out the sensitive areas and reduce redundant information; then, the CARS and GA algorithms were applied to identify the final sensitive wavelengths. This study demonstrated the sensitivity of chlorophyll to green peak, red valleys, red edges, and parts of the near-infrared region (750–920 nm) of the spectrum. Furthermore, Zhang et al. [15] provided a hybrid feature extraction method named "2DCOS." The 2D refers to hyperspectral amplitude (i.e., reflectance) and curve shape characteristics. Also, the curve shape characteristic was described by calculating the inverse tangent function between adjacent bands.

To summarize the issues mentioned above, sufficient spectral data mining with low data redundancy is urgently required to extract complete spectral features for accurate LCC estimation. Therefore, this article proposes a novel hybrid feature extraction method. First, the entire spectrum is divided into several areas based on the spectral response of chlorophyll and nitrogen. Second, segmental fitting and integral calculations are performed on the spectral curves of each area. Finally, the integral values from each area serve as inputs for subsequent inversion modeling. In this way, it is possible to obtain feature variables that simultaneously contain spectral reflectance and curve shape characteristics, while also achieving data dimension reduction.

The rest of this article is organized as follows. Section II introduces the deployment of field experiments, the acquisition of UAV-borne HRS images and LCC data, and the methodologies for constructing spectral features and building LCC inversion models. Section III presents the results of feature construction, validation of LCC inversion models, and comparisons with other feature extraction methods. Section IV provides an analysis of the obtained results. This article concludes in Section V with a summary of the findings.

II. Materials and Methods

A. Experimental Description

The study areas for the field experiments were located in Midu county (MD, red soil; altitude 1690 m; annual average precipitation of 824.4 mm; and annual average temperature of 17.3 °C) and Xiangyun county (XY, yellow soil; altitude 1840 m; annual average precipitation of 810.8 mm; and annual average temperature of 14.7 °C), within the Dali Bai Autonomous Prefecture (25°41′24″N, 100°11′24″E), Yunnan Province, Southwest China. The field experiments were conducted from May to September in both 2022 and 2023, involving four flue-cured tobacco cultivars: Hongda (Hd), disease-resistant Hongda (drHd), Yunyan 87 (Y87), and

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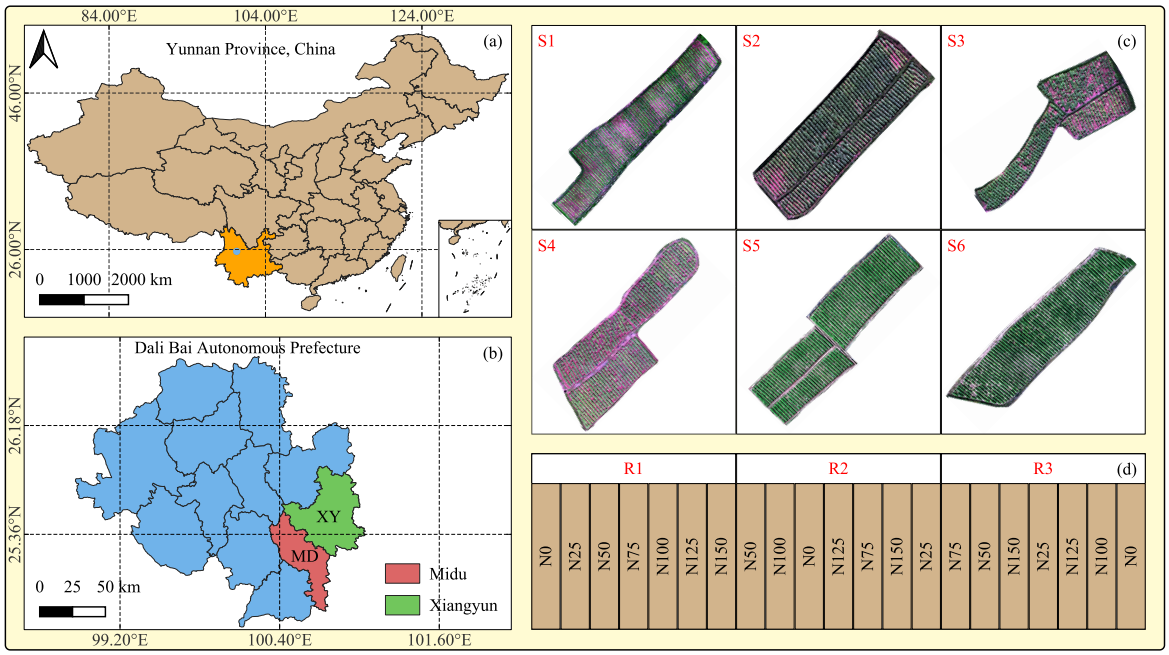


Fig. 1. Location of study areas and experimental plots. (a) Location of study areas in China. (b) Distribution of Midu and Xiangyun counties. (c) Employed six tobacco fields. (d) Design of experimental plots.

Yunyan 121 (Y121). These varieties were the main cultivars in the local area. Among them, Hongda and Yunyan exhibit significant differences in terms of plant geometric characteristics and canopy structure. The experiments employed a completely randomized block design, with six fields in 2022 (labeled S1–S6) and five fields in 2023 (S1 and S3–S6). The layout of the study area is shown in Fig. 1. Each field, approximately 1000 m^2, was set up with three replicates (labeled R1–R3), and within each replicate, seven plots were arranged (labeled N0–N150), each receiving a different gradients of nitrogen fertilizer treatment. Herein, N100 indicated the regular application rate for the fields, approximately 30 g/m^2, N0 indicated no fertilization, and N150 was 1.5 times the amount of N100; the rest could be deduced by analogy. The specific fertilization details can be found in Table I. Apart from the above arrangements, all other tobacco cultivation and management practices were maintained consistent with local customs.

TABLE I
DESIGN OF EXPERIMENTAL PLOT

Year	ID	Site	Specie	Fertilization (g/m^2)			
				1	2	3	4
2022	S1	MD	Hd	45	12	22.5	12
	S2	MD	Hd	7.5	18	37.5	0
	S3	MD	Hd	7.5	18	37.5	0
	S4	MD	Hd	15	22.5	30	0
	S5	XY	Y87	15	22.5	30	0
	S6	XY	Y121	15	22.5	22.5	0
2023	S1	MD	drHd	30	8	15	8
	S3	MD	drHd	5	12	25	0
	S4	MD	drHd	10	15	20	0
	S5	XY	Y87	10	15	20	0
	S6	XY	Y121	10	15	15	0

- S1-S3: N:P_2O_5:K_2O = 12:10:25;
- S4-S6: N:P_2O_5:K_2O = 10:10:24.

B. Spectral Image Acquisition and Preprocessing

UAV-borne HRS images were captured using the DJI Matrice 600 Pro[1] (Shenzhen DJI Technology Company Ltd., China) and the GaiaSky-mini2-VN[1] push broom imager (Jiangsu Dualix Spectral Imaging Technology Company Ltd., China). The obtained hyperspectral images contain 175 bands with wavelengths between 400 and 1000 nm. To ensure optimal imaging quality, the measurements were always carried out between 10:00 and 15:00 local time (GMT + 8) under a clear sky and windless conditions during the critical growing season, at an altitude of 80 m above the ground. Before the UAV took off, a gray calibration cloth was utilized for baseline measurement. Measurements were conducted three times in 2022 (labeled T1–T3) and twice in 2023 (labeled T4 and T5).

In this study, the average reflectance of pure tobacco pixels within each plot was extracted. The excess green index (ExG) method was applied in ENVI 5.6 (Exelis Visual Information Solutions Inc., Boulder, CO, USA) to eliminate nontobacco pixels, such as soil, shadows, and weeds. As demonstrated in our previous study, the classification effect of ExG method was highly significant [48]. The raw spectral curves were smoothed with the Savitzky–Golay filter, after which the first-order derivatives of the smoothed curves were calculated for subsequent operations.

C. Chlorophyll Content Measurement

The measurement of tobacco LCC data was conducted simultaneously with the acquisition of UAV-borne

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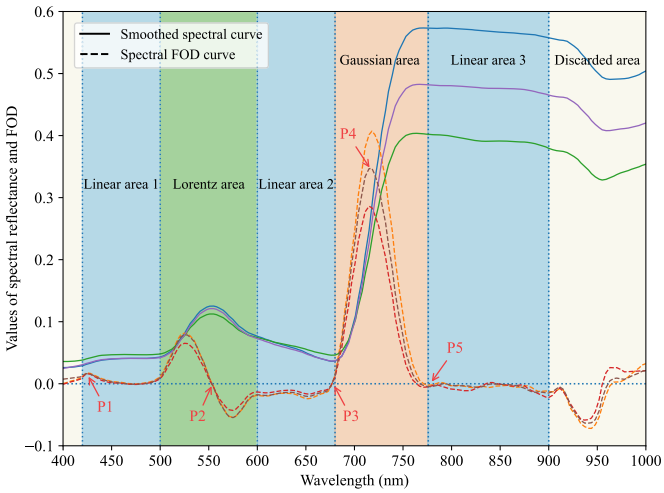


Fig. 2. Division of the entire spectral curve. For clear display, the first-order derivative data has been magnified tenfold.

hyperspectral images. The relative chlorophyll content of tobacco leaves was measured using an SPAD-502 Plus chlorophyll meter (Konica Minolta Sensing Inc., Osaka, Japan).[2] In each plot, five middle-positioned leaves from five representative tobacco plants were selected for measurement. SPAD readings were taken from five points on the fully expanded leaves (i.e., top middle, top left, bottom left, top right, and bottom right). The average of all 25 SPAD readings was then calculated as the final results for related plot. In addition, due to disease and waterlogging, 42 data from T1S3 and T5S4 were deemed invalid.

D. Spectral Feature Construction

As shown in Fig. 2, the entire spectral curve was roughly divided into several areas according to the spectral absorption and reflection characteristics of chlorophyll. Among them, the spectral areas affected by water (after 900 nm) and the noise parts at both ends of the spectral curve have been discarded. The remaining areas, i.e., three linear areas, a Lorentz area, and a Gaussian area, indicated that we will use linear, Lorentz, and inverse Gaussian functions to fit the curves in these areas. Herein, the inverse Gaussian function is commonly used to approximate the red edge region of the vegetation spectrum [49], [50]. Also, the Lorentz function is commonly used in physics to describe resonance phenomena, such as the shape of spectral lines and the propagation of electromagnetic waves [51]. The Lorentz function is characterized by a sharp peak, with smooth declines on both sides, and is notably long-tailed. These characteristics are similar to the green peak region spectrum. The fundamental formulas of Lorentz and inverse Gaussian functions are as follows:

$$L(\lambda) = \frac{1}{\pi}\frac{\gamma}{(\lambda-\lambda_l)^2+\gamma^2} \tag{1}$$

$$G(\lambda) = R_s - (R_s - R_g)\exp\left(\frac{-(\lambda-\lambda_g)^2}{2\sigma^2}\right) \tag{2}$$

[2]The subsequent LCC values stated in this article are SPAD readings.

where λ represents the wavelength, $L(\lambda)$ is the value of the Lorentz function at λ, and λ_l is the position of the curve peak; the parameter γ denotes the full-width at half-maximum. $G(\lambda)$ is the value of the inverse Gaussian function at λ, and R_g represents the minimum reflectance value in the red light region of the chlorophyll absorption zone, corresponding to the band λ_g. R_s is the maximum shoulder reflection in the near-infrared region, and σ is the standard deviation coefficient of the Gaussian function.

To ensure fitting accuracy and control fitting intervals, some special values of the first-order derivative (FOD) data were used to perform parameter initialization. Specifically, (λ_{P1}, $R_{\lambda_{P1}}$), (λ_{P3}, $R_{\lambda_{P3}}$), and (λ_{P5}, $R_{\lambda_{P5}}$) were used as initial values of three linear functions, respectively. (λ_{P2}, $R_{\lambda_{P2}}$) corresponded to (λ_l, $L(\lambda_l)$). (λ_{P3}, $R_{\lambda_{P3}}$) and (λ_{P5}, $R_{\lambda_{P5}}$) were used as the initial values of inverse Gaussian function. Furthermore, the fitting intervals of five functions were λ_{P1}–500, 500–600, 600–λ_{P3}, λ_{P3}–λ_{P5}, and λ_{P5}–900 nm, respectively. Moreover, the value of σ could be calculated by $\lambda_{P4} - \lambda_{P3}$. The integral of the spectral curves for each area could be calculated after the intervals have been determined. The integral expressions for the Lorentz function (labeled I_L) and the inverse Gaussian function (labeled I_G) are as follows:

$$\begin{aligned} I_L &= \int_{500}^{600}\frac{1}{\pi}\frac{\gamma}{(\lambda-\lambda_l)^2+\gamma^2}d\lambda \\ &= \frac{1}{\pi}\left[\arctan\left(\frac{600-\lambda_l}{\gamma}\right) - \arctan\left(\frac{500-\lambda_l}{\gamma}\right)\right] \end{aligned} \tag{3}$$

$$\begin{aligned} I_G &= \int_{\lambda_{P3}}^{\lambda_{P5}}\left[R_s - (R_s-R_g)\exp\left(\frac{-(\lambda-\lambda_g)^2}{2\sigma^2}\right)\right]d\lambda \\ &= R_s(\lambda_{P5}-\lambda_{P3}) - \frac{\sqrt{2\pi}\sigma(R_s-R_g)}{2} \\ &\quad\times\left[\mathrm{erf}\left(\frac{\lambda_{P5}-\lambda_g}{\sqrt{2}\sigma}\right) - \mathrm{erf}\left(\frac{\lambda_{P3}-\lambda_g}{\sqrt{2}\sigma}\right)\right] \end{aligned} \tag{4}$$

where "erf" is the abbreviation for error function. The formula is as follows:

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}dt. \tag{5}$$

By leveraging the well-fitted functions, the integral calculation of the spectral curves for each area has been simplified. In addition, the final constructed feature variable can be expressed as follows:

$$F_{\text{int}} = \{I_{l1}, I_L, I_{l2}, I_G, I_{l3}\} \tag{6}$$

where I_L and I_G denote the integral values of the Lorentz and Gaussian areas, respectively. Similarly, I_{l1}, I_{l2}, and I_{l3} denote the integral values of linear areas 1–3, respectively.

So far, we have completed the segmental integral of the entire spectral curve and constructed an eigenvariable with five eigenvalues, thereby preparing for the subsequent LCC model development. For convenience, we have named this feature construction method segmented fitting for integral (SFI).

E. Modeling and Evaluation

According to the existing literature, the ensemble learning-based methods typically perform better when the

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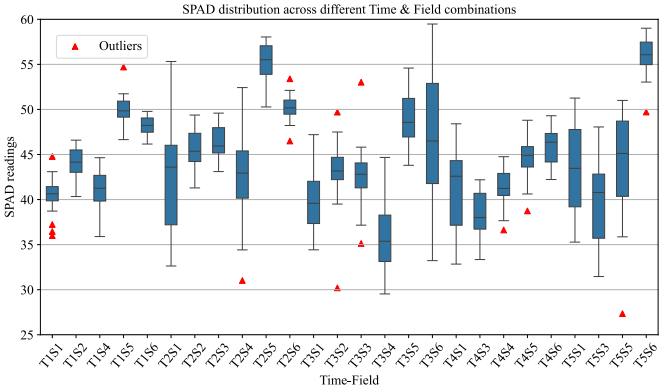


Fig. 3. Boxplot analysis for each site from T1 to T5 (T1S3 and T5S4 are discarded).

dataset consists of samples from different regions, species, and growing environments [25], [52], [53]. Therefore, this article selected two prevalent and commonly used ensemble learning algorithms, random forest (RF) and adaptive boosting (AdaBoost), to establish the LCC inversion model. For more detailed discussion on the strategies, frameworks, and representative algorithms of ensemble learning, refer to our previous works [48].

To compare the performance of models established using different feature extraction methods, this study employed three typical data dimensionality reduction algorithms for modeling, i.e., SPA, recursive feature elimination (RFE), and principal component analysis (PCA). Among them, SPA is a forward feature construction method that aims to optimize the feature space by iteratively selecting subsets of features that maximize orthogonality. In contrast, RFE is a backward feature elimination method that filters features by repetitively constructing a model and removing the least important features. As for PCA, it uses the linear transforms to reduce the original n-dimensional dataset to a new m-dimensional ($m \ll n$) dataset, named principal components (PCs). In this way, PCA retains the important information of the original data while effectively reducing the data dimensionality.

This study employed three quantitative indicators to evaluate the performance of models: coefficient of determination (R^2), root-mean-square error (RMSE), and mean absolute percentage error (MAPE). The specific calculation formulas of R^2, RMSE, and MAPE are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \overline{y}\right)^2} \tag{7}$$

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}{n}} \tag{8}$$

$$\mathrm{MAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\frac{\left|y_i - \widehat{y}_i\right|}{y_i} \tag{9}$$

where n is the number of samples, y_i is the measured SPAD value of the ith sample, $\widehat{y}_i$ is the predicted LCC of the ith sample, and $\overline{y}$ is the average of the measured LCC, respectively. A well-performing model necessarily has a high R^2 and lower values of RMSE and MAPE.

TABLE II
STATISTICAL RESULTS OF CHLOROPHYLL CONTENT

Dataset	Number	Max	Min	Avg	Std	CV (%)
Total	530	59.48	29.53	44.68	5.70	12.75
Training	371	59.48	30.19	44.94	5.71	12.71
Validation	159	57.78	29.53	44.10	5.64	12.78

• Maximum (Max), Minimum (Min), Average (Ave), Standard deviation (Std), Coefficient of variation (CV).

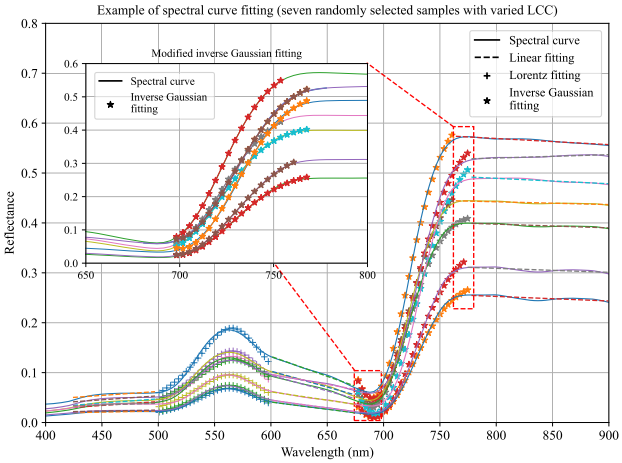


Fig. 4. Segmental fitting and local modification of the spectral curves.

III. RESULTS

A. Chlorophyll Statistical Analysis

First, the boxplot method was employed to identify outliers in the LCC values. Fig. 3 shows the SPAD distribution across different time-plot combinations, identifying a total of 16 outliers. Thus, the final number of valid samples in the total dataset was reduced to 530. To facilitate modeling and evaluation, the total dataset was split into training and validation sets in a 7:3 ratio using a Python (https://www.python.org) package. Table II shows the statistical information of LCC about each dataset. The minimal differences and high similarity among the datasets suggest that the division was reasonable.

B. Spectral Feature Extraction Results

Fig. 4 shows the fitting results of the spectral curves from seven randomly selected samples with different LCC values, and the fitting results around λ_{P3} and λ_{P5} were obviously not satisfactory. Thus, the fitting interval of the inverse Gaussian function was adjusted from λ_{P3}–λ_{P5} to $(\lambda_{P3}+20)-(\lambda_{P5}-10)$ nm. The fitting results of the modified inverse Gaussian function are shown in the top-left subfigure, which was significantly better than the unmodified results.

Fig. 5 shows the feature extraction results of SPA, RFE, and PCA. For both SPA and RFE, the relevant indices (RMSE and R^2) tended to converge when the number of screened bands reached a certain value. Specifically, the SPA-screened 19 bands (466, 512, 565, 578, 625, 646, 663, 676, 687, 697, 708, 721, 739, 750, 767, 792, 803, 882, and 900 nm) were relatively evenly distributed across the entire spectral curve. Also, the RFE-screened 16 bands (424, 440, 502, 531, 538, 565, 568, 575, 581, 588, 676, 725, 831, 838, 886, and

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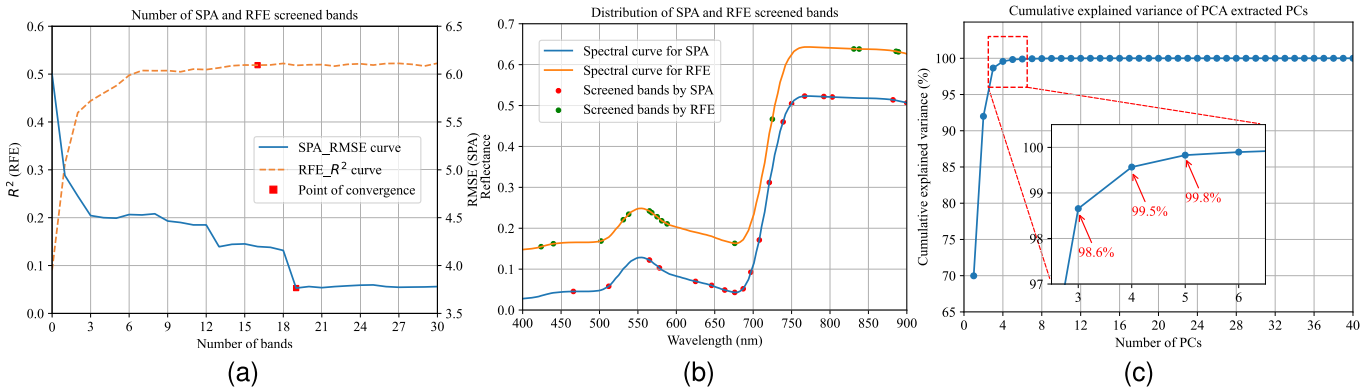


Fig. 5. Feature extraction results of SPA, RFE, and PCA. (a) Number of SPA- and RFE-screened bands. (b) Distribution of SPA- and RFE-screened bands. (c) Cumulative explained variance of PCA-extracted PCs.

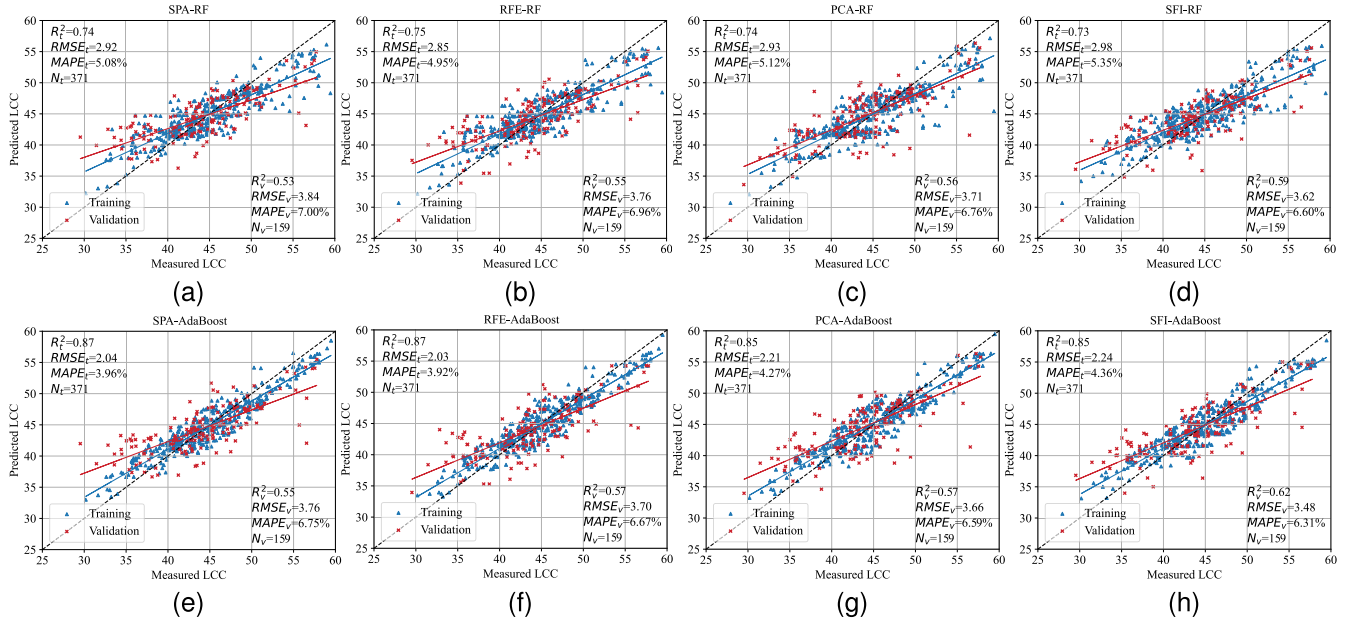


Fig. 6. Results of eight models with four feature extraction methods (SPA, RFE, PCA, and SFI) and two modeling methods (RF and AdaBoost). (a) SPA-RF. (b) RFE-RF. (c) PCA-RF. (d) SFI-RF. (e)SPA-AdaBoost. (f) RFE-AdaBoost. (g) PCA-AdaBoost. (h) SFI-AdaBoost.

889 nm) were mainly concentrated in the green peak (500–600 nm) and near-infrared (780–900 nm) regions. Different from SPA and RFE, the PCA achieves data dimensionality reduction by linear transformation. The cumulative explainable variance indicates the amount of information retained from the original data. Fig. 5(c) shows that when the number of PCs was 3–5, the retained information content reached 98.6%, 99.5%, and 99.8%, respectively. The percentage of explained variance for the first five PCs was 69.98%, 22.01%, 6.66%, 0.91%, and 0.26%. It can be deduced that the amount of information retained from the fifth principal component onward was already minimal. Therefore, the first four PCs were selected as the input variables.

TABLE III
SUMMARY OF EIGHT MODELS

Model	Training			Validation		
	R_t^2	RMSE$_t$	MAPE$_t$	R_v^2	RMSE$_v$	MAPE$_v$
SPA-RF	0.74	2.92	5.08	0.53	3.84	7.00
RFE-RF	0.75	2.85	4.95	0.55	3.76	6.96
PCA-RF	0.74	2.93	5.12	0.56	3.71	6.76
SFI-RF	0.73	2.98	5.35	**0.59**	**3.62**	**6.60**
SPA-Ada	0.87	2.04	3.96	0.55	3.76	6.75
RFE-Ada	0.87	2.03	3.92	0.57	3.70	6.67
PCA-Ada	0.85	2.21	4.27	0.57	3.66	6.59
SFI-Ada	0.85	2.24	4.36	**0.62**	**3.48**	**6.31**

• Adaboost (Ada), the unit of MAPE is %.

C. LCC Inversion Modeling and Evaluation

As mentioned above, we have established a total of eight models with four feature extraction methods (SFI, SPA, RFE, and PCA) and two modeling methods (RF and AdaBoost). The results are shown in Table III, and the related scatter plots are presented in Fig. 6. For the same modeling method, the models performed very similarly on the training set, but the R_v^2 values on the validation set were quiet different. The SPA-based models had the lowest validation performance. The evaluation indicators (R_v^2, RMSE$_v$, and MAPE$_v$) for models SPA-RF and SPA-AdaBoost were (0.53, 3.84, 7.00) and (0.55, 3.76, 6.75), respectively. The RFE-based models performed slightly better than the SPA-based models, and there was a 0.02 improvement on R_v^2. The performance of PCA-based models also showed improvement over the RFE-based models. PCA-RF outperformed RFE-RF, achieving a 0.01 increase in R_v^2 and along with reductions of 0.05 in RMSE$_v$ and

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0.2 in $MAPE_v$. R_v^2 of PCA-AdaBoost was the same with RFE-AdaBoost, but with slightly lower $RMSE_v$ and $MAPE_v$ values. Finally, the models developed using the SFI method all achieved optimal performance. The values of (R_v^2, $RMSE_v$, and $MAPE_v$) for SFI-RF and SFI-AdaBoost were (0.59, 3.62, 6.60) and (0.62, 3.48, 6.31), respectively.

IV. Discussion

A. Analysis of Modeling Results

The results of feature extraction and modeling suggest that the information entropy of the features can also significantly affect the final performance of a model. Herein, the concept of information entropy is derived from information theory, which refers to the effective information content of the data after removing redundancy [54]. From a qualitative perspective, features constructed by SFI method have the highest information entropy, as they contain nearly all spectral reflectance and curve shape characteristics. The features obtained by the other three methods do not contain spectral curve shape characteristics. Next are the PCA-extracted features, which retain at least 99.5% of the spectral reflectance characteristics. As for the RFE and SPA, both methods can extract a key part of the spectral reflectance characteristics. The difference is that RFE internally uses RF as the evaluator to calculate the importance of each band and then removes the least important bands. This provides a global view that SPA lacks. Since SPA requires selecting an initial band and then projecting this band onto others, bands are filtered by comparing the magnitude of the projection vectors. This greedy strategy-based selection process often leads to locally optimal solutions [42].

In the modeling phase, the commonly used algorithms RF and AdaBoost were selected to compare the performance of different feature extraction methods. The parameter settings were kept consistent across all models. Specifically, the parameters *n_estimators* and *max_depth* were set to [20, 120] and [3, 5], respectively. The optimal combination of parameters was searched by the GridSearch method in the scikit-learn package (https://scikit-learn.org). Although each model performed well in terms of $RMSE_v$ and $MAPE_v$, the value of R_v^2 was not very high. In addition, the difference between the maximum and minimum $RMSE_v$ values was 0.36, while for $MAPE_v$, it was 0.69. The primary reason is that the LCC values in the dataset are relatively concentrated, with 80.75% (428) of the LCC data distributed between 35 and 50, as shown in Fig. 3. Furthermore, some extreme values in the dataset can have a significant impact on the calculation of R^2, but due to their limited number, they will not significantly affect the RMSE and MAPE.

B. Stability Analysis of SFI

Different from the traditional VI-based feature constructed methods [31], [34], [36], the hyperspectral reflectance used in this study was extracted from pure tobacco pixels, making our SFI method independent of leaf surface structure and soil background. Compared with machine learning-based feature extraction methods [43], [44], there was no internal evaluator in SFI and only a set of position parameters for determining

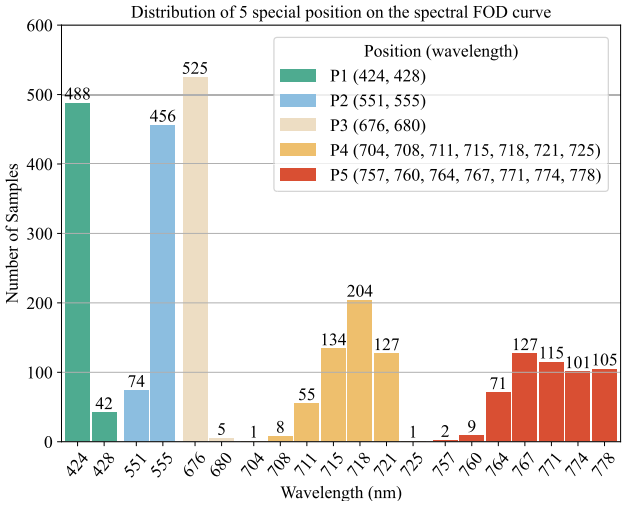


Fig. 7. Distribution of bands where P1–P5 are located.

TABLE IV
Statistical Modeling Results Under Different Intervals

Area	Interval (nm)	R_v^2	$RMSE_v$	$MAPE_v$
Linear area 1	λ_{P1}-480	0.58	3.63	6.60
	λ_{P1}-490	0.59	3.59	6.52
	λ_{P1}-510	0.59	3.60	6.52
	λ_{P1}-520	0.59	3.60	6.46
Linear area 2	600-($\lambda_{P3}-10$)	0.59	3.61	6.59
	600-($\lambda_{P3}-20$)	0.60	3.53	6.39
	600-($\lambda_{P3}+10$)	0.61	3.48	6.31
	600-($\lambda_{P3}+20$)	0.58	3.66	6.53
Linear area 3	λ_{P5}-880	0.58	3.64	6.62
	λ_{P5}-890	0.59	3.60	6.54
	λ_{P5}-910	0.58	3.63	6.55
	λ_{P5}-920	0.60	3.57	6.40
Lorentz area	500-580	0.61	3.53	6.39
	500-590	0.61	3.49	6.25
	500-610	0.60	3.57	6.33
	500-620	0.58	3.64	6.48
Gaussian area	($\lambda_{P3}+20$)-($\lambda_{P5}-30$)	0.58	3.65	6.62
	($\lambda_{P3}+20$)-($\lambda_{P5}-20$)	0.57	3.70	6.67
	($\lambda_{P3}+20$)-λ_{P5}	0.53	3.84	6.92
	($\lambda_{P3}+20$)-($\lambda_{P5}+10$)	0.55	3.77	6.67

the fitting interval. As shown in Fig. 7, the bands where P1–P5 were located exhibit high stability. Each of P1–P3 was associated with two specific wavelengths, and each of P4 and P5 was associated with seven specific wavelengths. The sample data in this study came from two different years, during which various environmental conditions varied significantly, yet the positions of P1–P5 were nearly stable. This stability indicated that the positions of P1–P5 were mainly related to the internal structure of the tobacco plants, rather than external factors. In this case, the fitting intervals were also nearly fixed, because the integral calculations did not involve any extra parameters, thus reducing the uncertainty in feature construction.

To evaluate the parameter sensitivity of SFI, we used the control variable method to test the impact of each interval on the performance of the model (using the SFI-Ada model as an example). The adjustments to each interval and the modeling results are shown in Table IV. For the Lorenz function and the three linear functions, changes in the interval (expansion or

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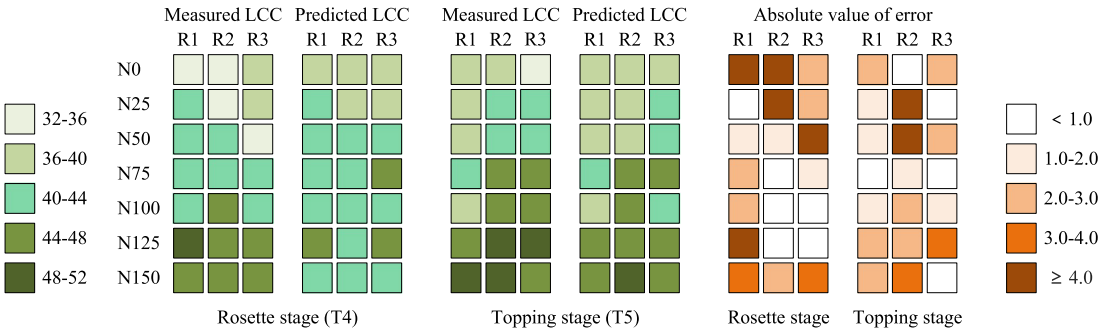


Fig. 8. Distribution and error maps of tobacco LCC at two stages.

contraction) had a small effect on the final model performance. As for the inverse Gaussian function, the contraction of the interval had a small effect on the model ($R_v^2 = 0.58$ and 0.57), whereas the expansion of the interval significantly affected the model performance ($R_v^2 = 0.53$ and 0.55). The reason might be that the extension of the interval has led to a decrease in fitting accuracy around λ_{P5}, which can be seen in Fig. 4. Thus, the optimal parameter selection strategy should be to choose the most appropriate fitting interval as much as possible to ensure the fitting accuracy of the spectral curve, especially the Gaussian area.

C. Potential for Practical Field Application

Taking S1 in 2023 as an example, the spectral data from 21 plots at each growth stage were input into the SFI-AdaBoost model to obtain the predicted LCC values for each plot, and then, the distribution and error maps of the measured and predicted LCC values were constructed. The distribution maps in Fig. 8 demonstrated a significant positive correlation between tobacco LCC and fertilization rate. The LCC at the topping stage was generally higher than at the rosette stage, primarily attributed to the increased leaf age. Comparing the distribution maps of predicted and measured LCC values across the two stages, the distribution of LCC had a high consistency. There were 10 and 14 plots where the predicted values agreed with the measured values at the rosette and topping stages, respectively. Meanwhile, the error map showed that there were 15 and 19 plots with prediction errors below 4.0 at the rosette and topping stages, respectively. In contrast, only five and two plots had prediction errors above 4.0 at these stage, all of which were situated within an adjacent gradient. Furthermore, the prediction accuracy was significantly higher at the topping stage compared with the rosette stage, with 81% of plots showing errors less than 3.0 at the topping stage, compared with 67% at the rosette stage. This suggested that the SFI-AdaBoost model could accurately predict the LCC of tobacco at the field scale, which can provide a valuable reference for farmers in fertilizer application and harvesting.

V. Conclusion

In this study, we used UAV-borne HRS data and LCC data (SPAD readings) of 530 tobacco samples collected from farmlands to develop chlorophyll inversion models. To improve the spectral utilization and stability of the feature variables, a novel feature construction method (SFI) based on segmented integral of spectral curves was proposed, which contains both spectral reflectance and curve shape characteristics. This method was fully analyzed, validated against field measured data, and compared with three different feature extraction methods, with main conclusions as follows.

1) According to the performance of the eight established models, the SFI-based models consistently outperformed others. Specifically, the SFI-AdaBoost model achieved the best performance with an R_v^2 of 0.62, an RMSE_v of 3.48, and an MAPE_v of 6.31%. The results confirmed the effectiveness of the SFI method in improving the accuracy of tobacco chlorophyll content prediction. Meanwhile, the stability analysis showed that the SFI method was less affected by the internal parameters and external environment.
2) The information entropy of the features also played an important role in model performance. However, it could not determine the model performance alone. In practice, accurate fitting is essential for the best modeling results. We should consider various factors comprehensively, such as the number of divided spectral areas, the fitting interval of each area, and the selection of fitting functions, which require more in-depth research in the future.
3) UAV-borne HRS technology is highly valuable for real-time and large-scale chlorophyll mapping, while SFI introduced a novel strategy for extracting stable spectral features from UAV-borne HRS data for chlorophyll modeling. The research in this study is meaningful for the application of UAVs in precision agriculture.

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